Technische Universität Wien

Algorithmics

course material: slides

Ostrich algorithm

14 languages

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In computer science, the **ostrich algorithm** is a strategy of ignoring potential problems on the basis that they may be exceedingly rare. It is named after the ostrich effect which is defined as "to stick one's head in the sand and pretend there is no problem". It is used when it appears the situation may be more cost-effectively managed by allowing the problem to continue to occur rather than to attempt its prevention.

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November 10, 2026

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CHAPTER 1

Computational Complexity (Cheat Sheet)

1.1 | P and NP

Def. 1 The class NP

There are two ways to define the complexity class NP:

1 | NP is the set of decision problems *solvable* in polynomial time by a *nondeterministic* Turing machine

2 | NP is the set of decision problems *verifiable* in polynomial time by a *deterministic* Turing machine:

- that is, $L \in$ NP if there are a polynomial p and a polynomial-time verifier V such that for every x:

$$x \in L \iff \exists u \text{ with } |u| \leq p(|x|) \text{ s.t. } V(x,u) = \text{accept}$$

Showing $L \in$ NP

1 | define the certificate u (the proposed solution)

2 | show that $|u| \leq$ a polynomial in the input length $|x|$ (NB: remember to count bits - n vertices take up $n \log n$ bits, not n)

3 | explain the mechanism of the verifier $V(x,u)$

4 | confirm that the running time of the verifier is polynomial in $|x|$

5 | prove both directions of correctness:

- explain why the verifier accepts a YES-instance
- explain why the verifier rejects a NO-instance (it cannot loop !)
  or, assume that the verifier accepts some certificate u, and prove that x must be a YES-instance

Def. 2 Karp Reduction

A polynomial-time Karp reduction translates an instance of problem A into an instance of problem B. Upper bounds transfer from B to A, meaning if $B \in P$, then $A \in P$. Conversely, hardness transfers from A to B (if $A \leq_p B$ and $A \notin P$, $B \notin P$).

Reductions are also transitive, meaning if $A \leq_p B$ and $B \leq_p C$, then $A \leq_p C$.

We have that $A \leq_p B$ iff there is a polynomial function f such that, for every string x:

$$x \in A \iff f(x) \in B$$

Showing $A \leq_p B$

1 | define the function $f(\text{instance of } A) = \text{instance of } B$, which must be computed from the instance alone

2 | state the size of $f(x)$ and the cost of computing it, in terms of $|x|$

3 | correctness ($\Longrightarrow$): prove that if x is a YES-instance of A, then $f(x)$ is a YES-instance of B (typically involves constructing a solution for $f(x)$ from a solution of x)

4 | correctness ($\Longleftarrow$): prove that if $f(x)$ is a YES-instance of B, then x is a YES-instance of A (typically involves recovering a solution for x from a solution of $f(x)$)

or, do the contrapositive: if x is a NO-instance of A, then $f(x)$ is a NO-instance of B

The core reductions covered in the course are:

- **Logic to logic:** SAT $\leq_p$ NAE-SAT
  - *Reduction by symmetry breaking:* pick a fresh variable s and append it to every clause: $(\ell_1 \vee \ldots \vee \ell_r) \mapsto (\ell_1, \ldots, \ell_r, s)$.
    - NAE has flip symmetry: if β satisfies, flipping all values β' also satisfies.
    - the shared s acts as a reference anchor: without loss of generality, take the solution where $s =$ false.
    - since s guarantees the required false in every clause, the clause is NAE-satisfied if and only if at least one original literal ℓ_i is true (which is the standard SAT condition)
- **logic to graphs:** 3-SAT $\leq_p$ CLIQUE
  - for a formula with m clauses:
    - create 3 vertices per clause (one per literal occurrence), put an edge between two vertices iff they are in different clauses and consistent ($u \neq \neg v$), set target clique size $k = m$
    - since there are no edges within the same clause, a clique of size m must pick exactly one literal per clause; because edges only connect non-contradictory literals, picking an m-clique never selects both x_i and $\neg x_i$, directly defining a valid satisfying truth assignment
- **graph to graph:** CLIQUE $\leq_p$ INDEPENDENT SET $\leq_p$ VERTEX COVER
  - CLIQUE $\leq_p$ IS:
    - *map:* $(G, k) \mapsto (\overline{G}, k)$, where $\overline{G}$ is the complement graph
    - a subset $S \subseteq V$ forms a clique in G (all pairs connected) $\iff$ S is an independent set in $\overline{G}$ (no pairs connected)
  - IS $\leq_p$ VC:
    - *map:* $(G, k) \mapsto (G, |V| - k)$ (flip the target size)
    - S is an independent set $\iff$ $V \setminus S$ is a vertex cover: since no edge can have both endpoints in S, every edge must touch at least one vertex in $V \setminus S$; thus, an IS of size $\geq k$ exists $\iff$ a VC of size $\leq |V| - k$ exists

- **Logic to Coloring:** NAE-3-SAT $\leq_p$ 3-COLORING
  - *Construction / Gadgets:*
    - have a neutral vertex B (assigned color *base*; remaining two colors represent *true* and *false*).
    - for each variable x, connect x and $\neg x$ to each other and both to B (forming a triangle with B); for each clause (ℓ_1, ℓ_2, ℓ_3), create a fresh triangle u_1, u_2, u_3 and add edges (u_1, ℓ_1), (u_2, ℓ_2), and (u_3, ℓ_3)
    - connecting every literal to B forces all literals to take colors from {true, false}, with x and $\neg x$ having opposite colors.
    - the triangle $\{u_1, u_2, u_3\}$ requires all 3 colors.
      - if ℓ_1, ℓ_2, ℓ_3 are all equal (all true or all false), that color is blocked for all u_i, leaving only 2 available colors for a 3-cycle (impossible to 3-color).
      - if ℓ_1, ℓ_2, ℓ_3 are not all equal, the three colors {true, false, B} can always be validly assigned to $\{u_1, u_2, u_3\}$.

Def. 3 **Gadgets**

Often, pieces of the target problem (e.g., a subgraph) can be designed to simulate a specific component of the source problem (e.g., a variable or a clause)

1 | **choice gadget:** how do you represent a boolean variable making a choice (True/False)?
- *example:* in mapping to 3-COLORING, a variable x is represented by a pair of adjacent vertices (x and $\neg x$) that are both connected to a universal "Base" vertex; the Base takes one color, forcing x and $\neg x$ to take the remaining two colors (representing True and False)

2 | **constraint gadget:** how do you enforce the rules of the source problem (e.g., "at least one literal must be true")?
- *example:* build a small subgraph connecting the literal vertices of a clause to a target vertex; the subgraph's edges are designed so that the whole gadget is only valid (e.g., fully colorable) if the inputs satisfy the source problem's specific rule

1.2 | Space Classes (L and NL)

Def. 4 **The Classes L and NL**

Instead of bounding time, we bound the memory (space) used on the work tape. The input tape is read-only, and we only count the memory used on the read/write work tape.

- **L (log-space):** the set of decision problems solvable by a *deterministic* Turing machine using $O(\log |x|)$ space
- **NL (nondeterministic log-space):** the set of decision problems solvable by a *nondeterministic* Turing machine using $O(\log |x|)$ space

Savitch / Immerman-Szelepcsényi: NL = coNL.

Showing $L \in$ L or NL

To show a problem is in L or NL, define an algorithm and prove it uses logarithmic space

1 | write the algorithm as a numbered list.

2 | list exactly what the work tape needs to hold (e.g., counters, vertex indices, pointers).

3 | write out the exact bit size for each variable, and show that the sum of all bit sizes is bounded by $O(\log |x|)$

- *Note:* a vertex index out of n vertices takes $\lceil \log_2 n \rceil$ bits; a counter up to n takes $\lceil \log_2 n \rceil$ bits

Def. 5 Log-space Reductions ($A \leq_L B$)

A log-space reduction from A to B is a deterministic algorithm that computes a mapping function f, transforming an instance of A into an instance of B, using only $O(\log |x|)$ work-tape space.

Showing $A \leq_L B$

1 | define f(instance of A) = instance of B, computed deterministically.

2 | state the variables needed to compute the map and their bit sizes, showing the sum is $O(\log |x|)$. (Do not count the space used to write the output, only the work tape).

3 | correctness ($\Longrightarrow$): prove that if x is a YES-instance of A, then $f(x)$ is a YES-instance of B

4 | crrectness ($\Longleftarrow$): prove that if $f(x)$ is a YES-instance of B, then x is a YES-instance of A (or prove the contrapositive for NO-instances)

Problems in L

- **BALANCED PARENTHESES:** given a string over '(' and ')', are they properly balanced? (uses a counter to track the current depth).
- **PALINDROME:** given a string, is it equal to its reverse? (uses two pointers)
- **EVEN-DEGREE:** given an undirected graph, does every vertex have an even degree? (uses a counter to iterate over edges for each vertex)
- **UNDIRECTED PATH:** given an undirected graph and vertices s, t, is there a path from s to t? (the directed version is NL-complete !)

Problems in NL

- **PATH (NL-Complete):** given a directed graph and vertices s, t, is there a directed path from s to t? *(canonical NL-complete problem; NL-hardness proofs start by reducing from PATH, often using configuration graphs)*
- **STRONG-CONNECTIVITY:** given a directed graph, is there a directed path between every pair of vertices?

- **CYCLE-THROUGH:** given a directed graph and a vertex v, is there a directed cycle containing v?
- **2-SAT and 2-UNSAT:** is a given 2-CNF formula satisfiable/unsatisfiable? (building an implication graph and searching for a directed path from a literal x to $\neg x$ and vice versa)

1.3 | The polynomial hierarchy

The polynomial hierarchy (PH) generalizes the complexity classes P, NP and coNP by classifying computational problems based on the number of alternating quantifiers required to verify their solutions.

(In standard NP, a non-deterministic machine guesses one certificate and checks it in polynomial time. The polynomial hierarchy extends this by allowing multiple, alternating phases of guessing. Each phase represents a shift in the logical burden of proof - from "does there exist?" (∃) to "is it true for all?" (∀). The number of times this burden shifts dictates the problem's level in the hierarchy.)

The base of every problem in the polynomial hierarchy is the polynomial-time predicate R

Def. 6 **Polynomial-time predicate**

A **polynomial-time predicate** $R(x, u_1, \dots, u_k)$ is a condition on strings that a polynomial-time algorithm checks.

- u_i are the certificates (guesses or moves generated by the quantifiers)
- the length of every u_i needs to be bounded by some polynomial relative to the size of the input x ($|u_i| \leq p(|x|)$)
- R then evaluates the conditions and outputs true/false in polynomial time

The classes of PH depend on the quantifier structure of the problem:

Def. 7 **Quantifier block**

A block is a *maximal run of equal quantifiers*. Two quantifiers of the same kind in a row are one block, since a pair of strings is one string; the *quantifier level* counts blocks.

The class Σ_k

A language L belongs in the class Σ_k if its defining logical statement starts with an existential quantifier, and has exactly $k-1$ alternations.

- e.g. $x \in L \in \Sigma_3 \iff \exists u_1 \forall u_2 \exists u_3 : R(x, u_1, u_2, u_3)$.

- We have that $\Sigma_1 = \text{NP}$!

The class Π_k

A language L belongs in the class Π_k if its defining logical statement starts with a universal quantifier, and has exactly $k-1$ alternations.

- e.g. $x \in L \in \Pi_2 \iff \forall u_1 \exists u_2 : R(x, u_1, u_2)$.

Π_k is the complement of Σ_k): $\Pi_i = \{L : \bar{L} \in \Sigma_i\}$ (negating the satatement swaps every ∃ with a

∀)

- We have that $\Pi_1 = \text{coNP}$!

1.3.1 | The Hierarchy Ladder

We have that:

$$\Sigma_i \cup \Pi_i \subseteq \Sigma_{i+1} \cap \Pi_{i+1}$$

, which means that any language in Σ_i or Π_i is also contained within both classes of the next level up (Σ_{i+1} and Π_{i+1}). This is proven by adding a "dummy" quantifier.

If a language is defined by a polynomial-time predicate R, we can artificially introduce a new quantifier over a variable v that R completely ignores. Because the verifier never evaluates v, the truth value of the statement is unchanged, but the formula gains the extra alternation required to place it in a higher class.

- moving from Σ_1 to Σ_2 (∃∀):

  suppose $L \in \Sigma_1$ (NP), defined as $x \in L \iff \exists u : R(x, u)$.

  we can rewrite this as $x \in L \iff \exists u \forall v : R'(x, u, v)$, where R' executes $R(x, u)$ and ignores v - the problem is now technically in Σ_2

- moving from Σ_1 to Π_2 (∀∃):

  using the same $L \in \Sigma_1$, we can alternatively pad the front of the statement: $x \in L \iff \forall v \exists u : R'(x, u, v)$, where R' again ignores v - the problem is now technically in Π_2.

Through this padding mechanism, the lower classes are subsumed by the intersection of the higher classes.

1.3.2 | The Quantifier Form and Absorption Rule

The Quantifier Form

To define a language L using the quantifier form, it is necessary to provide the statement, sizes, predicate, and class:

1 | **statement**: explicitly state $x \in L \iff Q_1 u$ (what u is, its size) $Q_2 v$ (what v is, its size): $R(x, u, v)$, written in words (e.g., "there is a ...such that for every ...")

2 | **sizes**: state that each object is bounded by an explicit polynomial in $|x|$
(N.B: a condition that is part of the question, e.g. "the set has at least k vertices", is checked inside R; it is not a size bound.)

3 | **predicate**: state that R is decidable in polynomial time, providing the reason

4 | **class**: read off the prefix: ∃ → NP, ∀ → coNP, ∃∀ → Σ_2, ∀∃ → Π_2

Not every ∀ or ∃ statement creates a new hierarchy level:

The absorb rule

The absorb rule dictates that if the candidate space for a quantifier is small enough that it can be listed and checked in polynomial time (e.g., checking every node in a graph), that quantifier

is “absorbed” directly into the polynomial-time predicate R (and it is called a **loop**).

A quantifier only remains in the prefix - and thus pushes the problem to a higher complexity class - if it iterates over an exponentially large candidate space, e.g. all possible 2^n Boolean truth assignments

Example: INDEPENDENT SET. Consider the problem: *is there a set $S \subseteq V$ with $|S| \geq k$, such that there are no edges inside S?*

The logical statement is: *there exists* ($\exists$) a set $S \subseteq V$ with $|S| \geq k$ such that *for every pair* ($\forall$) $\{a, b\} \subseteq S$, $ab \notin E$.

- The $\exists$ block remains: thge candidate space for S is 2^n (all possible subsets of n vertices). Because this scales exponentially, a deterministic machine cannot check all possibilities in polynomial time: the quantifier block remains.
- The $\forall$ block is absorbed: once a specific candidate set S is handed to the verifier, the inner condition checks “for every pair”. The maximum number of vertex pairs in S is bounded by n^2. Therefore, the machine simply runs a nested loop over the pairs to check the adjacency list. The $\forall$ quantifier is **absorbed** into the deterministic polynomial-time predicate R.

Therefore, the problem is in $\Sigma_1 = \text{NP}$.

To understand complete problems for the polynomial hierarchy, we use Quantified Boolean Formulas, which are the natural extension of the standard Boolean satisfiability problem (SAT).

Def. 8

A quantified Boolean formula is typically written in prenex form (meaning all quantifiers are pulled to the front) followed by a quantifier-free Boolean formula called the matrix (ψ).

$$Q_1 x_1 Q_2 x_2 \ldots Q_q x_q [\psi]$$

- where each Q_i is a quantifier ($\exists$ or $\forall$)

The formula is fully quantified, meaning every variable in the matrix ψ is bound by a quantifier (so the entire formula is either true or false).

Just as SAT is complete for NP, specific formats of QBFs act as the canonical complete problems for every level of the hierarchy.

- Σ_2-SAT: “given a formula $\exists X \forall Y [\psi]$, is it true?” is a complete problem for Σ_2.

  the matrix ψ must be an unrestricted Boolean formula: if ψ were in CNF, the $\forall Y$ check could be evaluated clause by clause, and the complexity would drop down to NP
- Π_2-SAT: “given a formula $\forall X \exists Y [\psi]$, is it true?” is Π_2-complete, and it remains complete even if ψ is in CNF

1.3.3 | The class DP

The class DP sits just above NP and coNP, but below the second level of the polynomial hierarchy ($\Sigma_2 \cap \Pi_2$). It captures problems that require solving an NP question and a coNP question simultaneously.

Def. 9 **DP**

A language belongs to DP if it is the intersection of two languages over the same instances: one from NP, and one from coNP:

$$\text{DP} = \{L_1 \cap L_2 : L_1 \in \text{NP}, L_2 \in \text{coNP}\}$$

N.B.: the class DP is different from NP ∩ coNP, which represents the class of single languages that have both a fast yes and a fast no verifier

- actually, DP = NP ∩ coNP if and only if NP = coNP

Showing $L \in$ DP

To show that a language L is in DP, you must split its defining condition into two independent halves.

For example: "exactly one solution" becomes "there is one" and "there are no two". "The optimum is exactly k" becomes "at least k" and "not at least $k+1$" (minimizing: "at most k" and "not at most $k-1$").

1 | **define the two halves**: explicitly define language L_1 (the "at least" or existence condition) and language L_2 (the "not more than" or uniqueness condition)

2 | **show the intersection**: formally state that a string $x \in L$ if and only if $x \in L_1 \cap L_2$

3 | **prove $L_1 \in$ NP**: define the certificate for L_1 (e.g., a solution of size k) and state that it is polynomially bounded and can be verified in polynomial time

4 | **prove $L_2 \in$ coNP** (usually by proving that $\overline{L_2}$ is in NP);

1.4 | The Class PSPACE

Def. 10 **PSPACE**

PSPACE is the class of decision problems solvable by a deterministic Turing machine using a polynomial amount of memory space.

Formally, it is defined as PSPACE = $\bigcup_{k\geq 1}$ SPACE($|x|^k$).

A fundamental property of space complexity (Savitch's Theorem) is that PSPACE = NPSPACE.

- While PH restricts problems to a constant, fixed number of quantifier alternations, PSPACE allows for an unbounded number of alternations (up to a polynomial limit). Consequently, the entire Polynomial Hierarchy is contained within PSPACE (PH ⊆ PSPACE).
- PSPACE captures the complexity of adversarial two-player games: any two-player game with board positions of polynomial size and a polynomially bounded maximum number of moves belongs in PSPACE. The nested alternating quantifiers (∃∀∃∀ ...) model the "if I move here, and you move there..." search required to determine a guaranteed winning strategy

The Complete Problem (TQBF): the problem of determining if a True Quantified Boolean Formula (TQBF) is true is PSPACE-complete. An instance of TQBF is a fully quantified Boolean formula

$Q_1 x_1 \dots Q_q x_q[\psi]$.

- unlike Σ_k-SAT, which is capped at k quantifier blocks, the number of quantifiers q in TQBF can grow linearly with the input size.

TQBF can be evaluated in polynomial space using a recursive algorithm. The algorithm evaluates each quantifier $Q_i x_i$ by testing both possible values of x_i (True and False). Because it can completely erase and reuse memory from the true branch when checking the false branch, the maximum memory required scales linearly with the recursion depth, keeping it in polynomial space.

1.5 | The Exponential Classes

These classes scale time and space bounds to the exponential level, dealing with problems where the input size allows for massive computational resources.

Def. 11 **The classes**

- EXP $= \bigcup_{k\geq 1} \text{TIME}(2^{|x|^k})$
- NEXP $= \bigcup_{k\geq 1} \text{NTIME}(2^{|x|^k})$

  kust as NP uses polynomially bounded certificates, NEXP can be viewed through a certificate lens: a language $L \in$ NEXP if there are certificates of exponential size (up to $2^{|x|^c}$) that can be verified by a deterministic algorithm in time polynomial to the *certificate's* size (which is exponential relative to the original input x)
- EXPSPACE $= \bigcup_{k\geq 1} \text{SPACE}(2^{|x|^k})$

A key concept at the exponential level is how data representation affects complexity. A **succinct** representation compresses an exponentially large structure into a small input.

- **succinct graphs:** instead of listing all edges of an explicit graph, a succinct graph uses a Boolean circuit C, which takes two b-bit vertex IDs and outputs 1 if there is an edge, and 0 otherwise

  because the input C is exponentially smaller than the explicit graph it represents, asking standard questions about the graph causes an exponential blowup in complexity:
  - a problem in P on an explicit graph becomes EXP-complete for a succinct graph
  - a problem in NP on an explicit graph (e.g., Hamiltonian Path) becomes NEXP-complete for a succinct graph

Padding Arguments: Upward Translation Padding is a proof technique demonstrating that a collapse at a lower complexity class triggers a collapse at a higher exponential class. The core mechanism involves artificially inflating the size of an input string to shift the problem into a lower complexity class relative to its new size.

The Padding Mechanism

1 | Consider a language $L \in$ NEXP

2 | Create a new "padded" language L' by appending dummy symbols to each input x until the new length is $N = 2^{|x|^k}$

3 | The original non-deterministic algorithm still takes N steps to evaluate the unpadded prefix x. Because N steps is polynomial relative to the massive new input size N, the padded language L' is structurally in NP

4 | If we assume P $=$ NP, then L' must be in P. To decide the original x, a deterministic machine simply writes out the pad (taking $2^{|x|^k}$ steps) and runs the P algorithm on L' (taking $2^{O(|x|^k)}$ steps). This proves $L \in$ EXP

Through padding, equalities carry upward, but inequalities carry downward via their contrapositive:

- P = NP $\implies$ EXP = NEXP.
- NP = coNP $\implies$ NEXP = coNEXP.
- EXP $\neq$ NEXP $\implies$ P $\neq$ NP

CHAPTER 2

A* Algorithm

2.1 | Informed Search

When trying to find the quickest or shortest route from a source node s to a target node t on a street network, uninformed search algorithms often aren't as fast as they could be.

For example, Dijkstra's algorithm is a classic uninformed search algorithm (that supports non-negative weights) that always examines the closest not-yet-examined node. On a simple grid, Dijkstra's expands uniformly in all directions, corresponding essentially to a Breadth-First-Search (BFS). While accurate, its simple implementation yields a time complexity of $O(|V|^2)$, which is typically too slow for large networks.

To speed up the calculation, we can exploit the properties of the graph by applying heuristic information to evaluate each reached node and always expand the most promising one.

2.1.1 | Greedy Best First Search

A purely heuristic approach evaluates nodes based on a function $h(x)$, which represents the "direct" (unobstructed) distance to the goal t. Common examples include the Euclidean distance $d_2(x,t)$ or the Manhattan distance $d_1(x,t)$.

Greedy Best First Search always expands the reached node with the minimum distance to the goal. While this minimizes the number of expanded nodes, it **does not guarantee an optimal solution**, as it can easily get trapped by obstacles or take suboptimal detours.

2.2 | The A* Algorithm

To avoid the inefficiencies of Dijkstra's algorithm and the mistakes of Greedy Best First Search, the **A* Algorithm** (Hart, Nilsson, Raphael, 1968) combines the advantages of both.

It utilizes a priority function $f(x)$ for all $x \in V$, defined as:

$$f(x) = g(x) + h(x)$$

Where:

- $g(x)$ is the best known cost so far to reach node x from the start node s (the "Dijkstra" part).
- $h(x)$ is the estimated cost from x to the goal t, also known as the "cost-to-go" (the "Greedy" part).

This way, $f(x)$ estimates the **total cost** of the path from s to t that passes through node x.

Implementation and Execution Details

During execution, A* maintains a priority queue of nodes to expand, sorted by their $f(x)$ value:

A* Algorithm

```
initialize g(s) = 0 and g(x) = ∞ for all x ∈ V \ {s}
insert source s into priority queue Q with priority f(s) = h(s)
while Q ≠ ∅ do
    extract node x with minimum f(x) from Q
    if x = t then
        return f(t) and the path (using predecessor list)
    end if
    for all outgoing edges (x, v) ∈ A do
        g' ← g(x) + w_xv
        if g' < g(v) then // found a new or better path to v
            g(v) ← g'
            pred(v) ← x
            insert or update node v in Q with priority f(v) = g' + h(v)
        end if
    end for
end while
return no path to target t
```

NB:

- the algorithm continues until the target node t is *selected for expansion* (extracted from the queue), not until it is seen
- A* may find a shorter path to a node that has already been expanded; in this case, the node gets its $g(x)$ updated and is re-inserted into the priority queue

2.3 | Properties of A*

Thm. 1 **[Termination]**

A* will always terminate in a limited time.

Proof

The inner loop performs a number of iterations proportional to the out-degree of the node ($O(|V|)$), which terminates. There are finitely many acyclic paths from s to other nodes in the graph, and A* only ever considers acyclic paths. On each major iteration, a new acyclic path is considered because a node is either expanded for the first time, or it is re-expanded via a shorter path. Thus, at worst, A* explores every acyclic path.

Thm. 2 **[Completeness]**

If a path from s to t exists, A* will always find one.

Proof

Similarly to DFS and BFS, after expanding a node x, A* has found paths from s to all nodes directly reachable via x. The only case where A* stops without finding a path to t is when no further reachable unexpanded nodes exist. This implies no path to t exists in the graph.

2.3.1 | Heuristics and Optimality

A* is not unconditionally guaranteed to find an optimal path. Its optimality depends entirely on the properties of the chosen heuristic $h(x)$.

Def. 12 **Admissible Heuristic**

Let $h^*(x)$ be the true minimal cost from x to t. A heuristic $h(x)$ is **admissible** if it represents a lower bound. Formally:

$$h(x) \leq h^*(x) \quad \forall x \in V$$

An admissible heuristic is always "optimistic" (it never overestimates the cost). $d_1(x,t)$, $d_2(x,t)$, and even $h(x) = 0$ are all admissible heuristics.

(it can be optimistic but not pessimistic because if it is optimistic you might just end up taking more time, but if it is pessimistic you might miss the shortest path (a pessimistic heuristic overestimates the cost, which can cause the algorithm to falsely assume the optimal path is too expensive and settle for a suboptimal route))

Thm. 3 **[Optimality]**

If $h(x)$ is admissible, A* is guaranteed to find a least-cost path.

Proof

Suppose A* finds a suboptimal path with cost $f' > f^* = h^*(s)$. There must exist a node x which is unexpanded and whose path from s is the beginning of a true shortest path to t. Since A* terminated, it must be that $f(x) \geq f'$. However, we know: $f(x) = g(x) + h(x) = g^*(x) + h(x) \leq g^*(x) + h^*(x) = f^*$ So, $f^* \geq f(x) \geq f'$, which contradicts the assumption that $f' > f^*$.

Def. 13 **Monotonic (Consistent) Heuristic**

A heuristic $h(x)$ is **monotonic** if it satisfies the triangle inequality for all edges $(x,y) \in A$, and the goal heuristic is zero:

$$h(x) \leq w_{xy} + h(y) \quad \text{and} \quad h(t) = 0$$

- monotonic heuristics ensure that the f-values along any path are non-decreasing ($f(x) \leq f(y)$)
- it is impossible to decrease $f(x)$ by extending a path
- this guarantees no re-expansions of nodes are ever necessary, making it comparable to Dijkstra (on non-negative weights)

> **N.B.**: monotonicity implies admissibility!

2.4 | Complexity and Variants

A* is optimally efficient with respect to the used heuristic $h(x)$, meaning no other complete algorithm using the exact same heuristic will expand fewer nodes.

- note that Dijkstra's algorithm is the special case of A* where $h(x) = 0$!

The runtime is generally $O(d^l)$ (where l is the number of nodes on the shortest path and d is a constant), but the actual performance relies heavily on the quality of $h(x)$.

> Heuristic Dominance and the 8-Puzzle
>
> A better heuristic leads to better A* performance. If $h_1(x) \leq h_2(x)$ for all $x \in V$, we say that h_2 **dominates** h_1. A* is guaranteed to expand no more nodes using h_2 than it would with h_1.
>
> With a nearly perfect heuristic $h(x) \approx h^*(x)$, runtime drops to $\Theta(l)$.
>
> For example, for the 8-Puzzle problem:
>
> - $h_1(x)$: number of tiles in wrong positions.
> - $h_2(x)$: sum of Manhattan distances of each tile to its target location.
>
> Because h_2 dominates h_1, running A* with h_2 reduces the number of expanded nodes compared to h_1 (e.g., 1,641 nodes vs 39,135 nodes for a 24-move solution).

2.4.1 | Improvements and Variants

For very complex problems, A* might still consume too much memory or time. Common improvements include:

- **Bidirectional search**: running search from s to t and from t to s simultaneously until they meet.
- **Weighted A***: "inflating" the heuristic (e.g., $f(x) = g(x) + W \cdot h(x)$) to place more emphasis on reaching the goal quickly.

  this sacrifices admissibility (optimality is no longer guaranteed), but it provides a bounded sub-optimality in exchange for big speedups
- **Preprocessing**: pre-calculating all-pairs shortest paths or storing pairwise minimum distances to create near-perfect heuristics
- **Hierarchical approach**: searching on abstraction levels (e.g., plan a route on major highways first, then refine to local main roads, and finally down to street level)
- **Iterated Deepening A* (IDA*)**: designed primarily to limit memory usage; it mimics Iterative Deepening DFS, performing a depth-first search while pruning any node where $f(x)$ exceeds a progressively increasing threshold c.

  while it guarantees finding the optimal solution with lower memory bounds, it is generally much slower than standard A* due to repeated work

CHAPTER 3

Planar Graphs

3.1 | Basic Concept

A graph $G = (V, E)$ is called planar if it can be drawn in the plane in a way such that no two edges intersect geometrically except at a vertex to which both are incident. Such a drawing is called a **planar embedding** of G, or a plane graph. A plane graph G divides the plane into connected regions called **faces**. The unbounded region is called the outer or external face of G.

3.2 | Euler's Formula

Thm. 4 **[Euler's formula]**

Let $G = (V, E)$ be a connected, planar graph, and let f denote the number of faces. Then

$$|V| - |E| + f = 2$$

holds for any planar embedding of G.

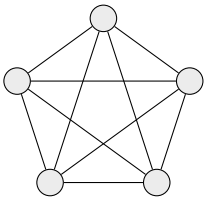
From Euler's formula, we can derive several important corollaries:

- if G is a simple, planar graph with $|V| \geq 3$, then the number of edges is bounded by $|E| \leq 3|V| - 6$
- for any simple, planar, bipartite graph with $|V| \geq 3$ vertices and $|E|$ edges, the bound is tighter: $|E| \leq 2|V| - 4$
- each simple, planar graph contains a vertex v of degree $d(v) \leq 5$

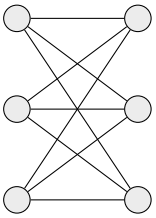
3.3 | Characterizations of Planar Graphs

Certain graphs cannot be drawn in the plane without crossing edges. The two fundamental examples are known as the Kuratowski graphs:

K_5



$K_{3,3}$



- K_5: the complete graph on 5 vertices

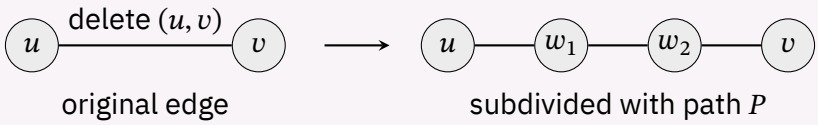
- $K_{3,3}$: the complete bipartite graph with two partitions of 3 vertices each

To establish whether a graph is planar, we define relationships to these Kuratowski graphs using subdivisions and minors.

Def. 14 **Subdivision**

Subdividing an edge $(u, v) \in E$ of a graph $G = (V, E)$ is the operation of deleting (u, v) and adding a path $P = (u, w_1, ..., w_k, v)$, $w_i \notin V$, $1 \leq i \leq k$ to G.

A graph G is called a subdivision of a graph G' if it is obtained by subdividing some of the edges of G'.

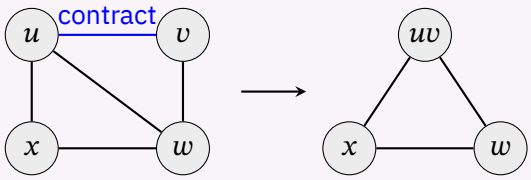


Thm. 5 **[[Kuratowski, 1930]]**

A graph is planar if and only if it does not contain a subdivision of K_5 or $K_{3,3}$.

Def. 15 **Graph Minor**

A graph H is a minor of G if H is obtained from G by deleting and contracting edges and/or deleting vertices.

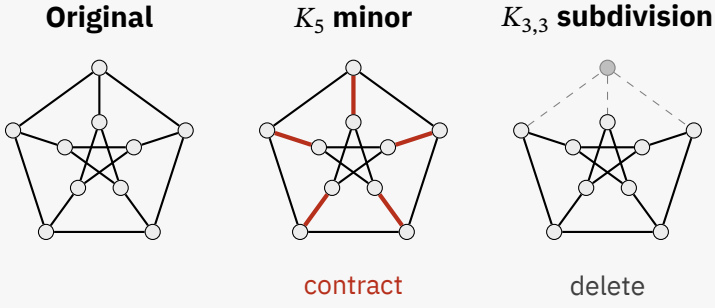


Thm. 6 **[[Wagner]]**

A graph is planar if and only if it has no minor isomorphic to K_5 or $K_{3,3}$.

Example: Peterson graph

The Peterson graph is a classic example of a non-planar graph.



- by contracting specific edges, we obtain the K_5 minor
- by deleting specific edges, we obtain a subdivision of $K_{3,3}$

Thus, we can prove its non-planarity with both Wagner's and Kuratowski's theorems.

3.4 | Linear Planarity Testing Algorithms

Several algorithmic approaches have been developed to test planarity and construct planar embeddings in linear time:

- **Path-Addition Approach**: the first linear-time planarity testing algorithm, introduced by Hopcroft and Tarjan (1974).

  starts from a cycle and adds one path at a time to the structure

  (Mehlhorn and Mutzel (1996) later clarified how to use this to construct the actual planar embedding)

- **Vertex-Addition Approach**: first considered by Lempel, Even, and Cederbaum (1967)

  adds vertices one-by-one following an st-numbering

  (Booth and Lueker (1976) proved it can be implemented in linear time using PQ-trees, and Chiba et al. (1985) showed how to construct the embedding)

- **Block Embedding Approach**: proposed by Shih and Hsu (1999) and Boyer and Myrvold (1999)

  first constructs a DFS-tree and then adds back edges to iteratively construct a planar embedding if one exists

  Boyer and Myrvold (2004) later introduced a new variant with a simpler implementation based on edge addition

CHAPTER 4

Randomized Algorithms

4.1 | Some Probability Theory

4.1.1 | The Basics

Def. 16 **Finite Probability Space**

- **sample space** Ω: set of all possible outcomes (e.g., all possible outcomes of rolling a dice)

  every point $i \in \Omega$ has a nonnegative probability $p(i)$, such that $\sum_{i \in \Omega} p(i) = 1$
- event $\mathcal{E}$: subset of the sample space ($\mathcal{E} \subseteq \Omega$).
- **probability of** $\mathcal{E}$:

$$Pr[\mathcal{E}] = \sum_{i \in \mathcal{E}} p(i)$$

- complement:

$$Not(\mathcal{E}) = \Omega - \mathcal{E}$$

Def. 17 **Conditional probability & independence**

- **conditional probability** of event $\mathcal{E}$ given event $\mathcal{F}$:

$$Pr[\mathcal{E}|\mathcal{F}] = \frac{Pr[\mathcal{E} \cap \mathcal{F}]}{Pr[\mathcal{F}]}$$

- **independence**:

  events $\mathcal{E}_1, \dots, \mathcal{E}_n$ are independent if and only if for all subsets $I \subseteq \{1, \dots, n\}$:

$$Pr\left[\bigcap_{i \in I} \mathcal{E}_i\right] = \prod_{i \in I} Pr[\mathcal{E}_i]$$

Union bound

For any events $\mathcal{E}_1, \dots, \mathcal{E}_n$, the probability of their union is bounded by the sum of their individual probabilities:

$$Pr\left[\bigcup_{i=1}^{n} \mathcal{E}_i\right] \leq \sum_{i=1}^{n} Pr[\mathcal{E}_i]$$

4.1.2 | Random Variables and Expected Values

A **random variable** $X : \Omega \to \mathbb{N}$ is a function from the sample space to the natural numbers.

The **expected value** (or expectation) of X is defined as:

$$E[X] = \sum_{j=0}^{\infty} j \cdot Pr[X = j]$$

example: waiting for a first success
Suppose a trial is successful with probability $0 < p < 1$ and fails with probability $1 - p$. If X is the number of independent trials until the first success, the expectation is:

$$E[X] = \sum_{j=0}^{\infty} j \cdot (1-p)^{j-1} p = \frac{p}{1-p} \sum_{j=0}^{\infty} j \cdot (1-p)^{j} = \frac{p}{1-p} \cdot \frac{1-p}{p^2} = \frac{1}{p}$$

Linearity of expectation

Let X and Y be random variables over the same probability space. Then:

$$E[X + Y] = E[X] + E[Y]$$

Example: guessing cards

Given a deck of n cards, repeatedly guess the top card before turning it over. Let X be the number of correctly guessed cards, and $X_i = 1$ if the i-th card is guessed correctly (0 otherwise).

- *memoryless*:

$$E[X_i] = \frac{1}{n} \quad \text{so} \quad E[X] = \sum_{i=1}^{n} \frac{1}{n} = 1$$

- *with memory*:

$$E[X_i] = \frac{1}{n-i+1} \quad \text{so} \quad E[X] = \sum_{i=1}^{n} \frac{1}{n-i+1} = \sum_{i=1}^{n} \frac{1}{i} = \mathcal{H}_n \approx \ln n$$

(where $\mathcal{H}_n$ is the n-th Harmonic number, satisfying $\ln(n+1) < \mathcal{H}_n \leq 1 + \ln n$).

Example: coupon collector's problem

Assume there are n types of coupons, and in each round, you get one at random (equally likely).

Let X be the number of rounds to collect all types.

We define *phase* j as the period where you have j types and are waiting for the $(j+1)$-th type. Let X_j be the number of rounds in phase j.

- the success probability in phase j is $p(j) = \frac{n-j}{n}$.
- the expected rounds for phase j is $E[X_j] = \frac{1}{p(j)} = \frac{n}{n-j}$.
- by linearity of expectation: $E[X] = \sum_{j=0}^{n-1} E[X_j] = n \sum_{j=0}^{n-1} \frac{1}{n-j} = n\,\mathcal{H}_n \approx n \ln n$.

4.2 | Randomized Algorithms

Randomized algorithms depend on **uniformly random numbers** as an **auxiliary input** to guide behaviour (these numbers are usually implemented by a pseudo-random number generator).

> Example: quicksort
>
> Deterministic quicksort has a worst-case runtime of $\Theta(n^2)$, when it gets handed specific input permutations (e.g., an already sorted array).
>
> We can randomise quicksort by selecting the pivot element uniformly at random instead of using a deterministic position(standard quicksort has a "fixed" pivot).
>
> This way, the worst case is still $\Theta(n^2)$, but it happens only if the internal random number generator repeatedly picks extreme values (very unlikely) (so, since choices are random, there are no bad input permutations that will always run in $\Theta(n^2)$).
>
> The expected runtime is $\Theta(n \log n)$ for all possible input permutations. The expectation is taken over all possible random decisions made by the algorithm, avoiding dependency on the input data distribution.

Randomized algorithms reduce runtime and/or memory usage for worst case input, and they avoid dependency on input data.

> Def. 18 **Monte Carlo vs Las Vegas**
>
> There are two types of randomized algorithms:
>
> - **Monte Carlo** algorithms: always **time-efficient**, but not always correct (correct with high probability)
> - **Las Vegas** algorithms: always **correct**, but not always time-efficient

4.2.1 | Randomized Primality Test

We are given a positive integer n, and we want to determine whether it is prime.

The first issue we run into is "bit complexity". Standard algorithm analysis often assumes arithmetic operations (like addition) take constant time ($O(1)$). However, this is only true for numbers small enough to fit in standard CPU registers (e.g., 64-bit integers).

For arbitrarily large numbers, addition takes time proportional to the length of the number. Therefore, we use **bit complexity**:

- represent n as a binary number with $k = \lfloor \log_2 n \rfloor + 1$ bits
- the number of fundamental bit operations (rather than high-level arithmetic operations) becomes the criterion for time complexity

For example, if we wanted to check primality via the "division method" - checking if n is divisible by 2 or by some odd number in $\{3, \dots, \lfloor \sqrt{n} \rfloor\}$ -, we would need to do $O(\sqrt{n})$ divisions, with a bit complexity of $\frac{O(2^k 2}{)}$ with $k = \Theta(\log n)$.

We need to use another method. We can make use of **Fermat's Little Theorem**, which states:

$$\text{if } n \text{ is prime, } a^{n-1} \equiv 1 \bmod n \text{ for all } a \in \{1, \dots, n-1\}$$

This allows us to say that, if for some a

- $a^{n-1} \not\equiv 1 \bmod n$, then n is definitely not prime (and a is a witness for n's compositeness)
- $a^{n-1} \equiv 1 \bmod n$ then n is prime with some probability

This gives us the following algorithm:

```
procedure FermatPrimalityTest(n, s)
    for i ← 1 to s do
        a ← random(2, n − 1)
        if Witness(a, n) then
            return COMPOSITE
        end if
    end for
    return PRIME
end procedure
```

Which essentially says that, if no witness for the number has been found after s iterations, then n is likely prime.

- this is a **Monte Carlo** algorithm (as it could return a false positive)

To implement this efficiently, we need a way to compute $a^{n-1} \bmod n$ in a cheap way.

> Efficient Modular Exponentiation
>
> We can break down the exponentiation using the binary representation of $n-1$: $(b_k, b_{k-1}, \dots, b_1)$.
>
> We then have that:
>
> - $a^{2c} \bmod n = (a^c \bmod n)^2 \bmod n$
> - $a^{c+1} \bmod n = (a^c \bmod n) \cdot a \bmod n$

This yields the following algorithm for the witness check:

```
procedure Witness(a, n)
    Let (b_k, ..., b_1) be the binary representation of n − 1
    result ← 1
    for i ← k down to 1 do
        result ← (result · result) mod n
        if b_i = 1 then
            result ← (result · a) mod n
        end if
    end for
    if result ≠ 1 then
        return true // a is a witness for n not being prime
    else
        return false // a is not a witness
    end if
end procedure
```

- multiplication and modulo of two k-bit numbers takes $O(k^2)$ time
- the "witness" function runs in $O(k^3)$

- the total running time of the Fermat algorithm is $O(s \cdot k^3)$

The test fails for *Carmichael numbers*, which are composite but incorrectly satisfy $a^{n-1} \equiv 1 \bmod n$ for all $a \in \{1, \dots, n-1\}$. Fortunately, these are extremely rare (e.g., there are only 246,683 below 10^{16}).

For all other odd, non-prime numbers $n > 2$:

- there are at least $\frac{n-1}{2}$ witnesses
- in each iteration, a randomly chosen a is a witness with probability $\geq \frac{1}{2}$
- after s iterations, the probability of not finding a witness is $\leq \frac{1}{2^s}$

4.3 | Analysis of (Randomized) Quicksort

The dominant cost of the algorithm is the comparison of elements to the pivot.

To analyse this, we conceptually sort the elements of the array as $x_1 < x_2 < \cdots < x_n$, and we define a variable $X_{i,j}$ s.t.

$$X_{i,j} = \begin{cases} 1 & \text{if } x_i \text{ is compared to } x_j \\ 0 & \text{otherwise} \end{cases}$$

Because the pivot is selected uniformly at random from the current subarray $A[l \dots r]$, we have:

$$Pr[\text{random}(l, r) = i] = \frac{1}{r - l + 1}$$

We can visualise the recursive pivot choices as a **binary search tree** (where the nodes are the randomly chosen pivot elements), in which every element is only compared with its direct ancestors or descendants.

Very importantly, if we consider two elements x_i and x_j, we know that they will only ever be compared if and only if *one of them is chosen as a pivot* before any element in between them is chosen: if any intermediate element x_k (where $i < k < j$) is selected as a pivot first, it would split them: x_i goes to the left subarray, x_j goes to the right subarray, and they would never be compared.

Probability of a single comparison

Consider the subset of elements $\{x_i, x_{i+1}, \dots, x_j\}$. This subset contains $j - i + 1$ elements.

Since pivots are chosen uniformly at random, every element in this subset is equally likely to be the first one chosen as a pivot. The only choices that result in a comparison between x_i and x_j are picking x_i or picking x_j.

Therefore, the probability that x_i and x_j are compared is exactly:

$$Pr[X_{i,j}] = \frac{2}{j - i + 1}$$

Expected total comparisons

We want to find the expected total number of comparisons, $E[C]$.

Using the Linearity of Expectation, we sum the expectations (which equal the probabilities) of all indicator variables:

$$E[C] = E\left[\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} X_{i,j}\right] = \sum_{i=1}^{n-1}\sum_{j=i+1}^{n} Pr[X_{i,j}] = \sum_{i=1}^{n-1}\sum_{j=i+1}^{n} \frac{2}{j-i+1}$$

We simplify this using a change of variables, letting $k = j - i + 1$:

$$E[C] = \sum_{i=1}^{n-1}\sum_{k=2}^{n-i+1} \frac{2}{k} \leq 2\sum_{i=1}^{n}\sum_{k=1}^{n} \frac{1}{k} = 2n\mathcal{H}_n$$

Where $\mathcal{H}_n = \sum_{k=1}^{n} \frac{1}{k}$ is the n-th Harmonic number. Since $\ln(n+1) < \mathcal{H}_n \leq 1 + \ln n$, we know $\mathcal{H}_n \approx \ln n$.

Therefore:

$$E[C] = E[T] = \Theta(n \log n)$$

4.4 | MAX 3-SAT

Def. 19 **MAX-3SAT Problem**

input: a set of clauses $C_1, \ldots, C_k$, each of length 3, over a set of binary variables $X = \{x_1, \ldots, x_n\}$.

goal: find a variable assignment satisfying as many clauses as possible.

> MAX-3SAT is NP-hard!

Randomized idea: set each variable independently to true with probability $\frac{1}{2}$ and to false otherwise.

Expected satisfied clauses

Given a MAX-3SAT instance with k clauses, the expected number of clauses satisfied by a random assignment is $\frac{7}{8}k$.

Proof

Define an indicator variable Z_j which equals 1 if clause C_j is satisfied, and 0 otherwise. Let $Z = \sum_{j=1}^{k} Z_j$ be the total number of satisfied clauses.

A clause of length 3 fails only if all 3 literals are false. Since each variable is chosen independently with probability $\frac{1}{2}$, the probability a clause fails is $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

Thus, $Pr[C_j \text{ is satisfied}] = 1 - \frac{1}{8} = \frac{7}{8}$.

By linearity of expectation:

$$E[Z] = \sum_{j=1}^{k} E[Z_j] = \sum_{j=1}^{k} Pr[C_j \text{ is satisfied}] = \frac{7}{8}k$$

Lower Bound on Satisfiable Clauses

For *every* instance of MAX-3SAT there is an assignment that satisfies $\geq \frac{7}{8}$ of all clauses.

Proof

As $E[Z] = \frac{7}{8}k$, the probability $Pr[Z \geq \frac{7}{8}k]$ of constructing such an assignment must be greater than 0. Consequently, such an assignment must exist.

> We can show the existence of a structure by providing a random construction process that succeeds with positive probability.

4.4.1 | Johnson's Algorithm

While the expectation is $\frac{7}{8}k$, a random variable might almost always be below its mean.

To have a reliable algorithm, we must determine the actual probability p of generating a good assignment.

Probability of Success

The probability p that a random assignment satisfies $\geq \frac{7}{8}k$ clauses is $\geq \frac{1}{8k}$.

Proof

Let p_j be the probability that exactly j clauses are satisfied.

Then $p = \sum_{j \geq 7k/8} p_j$.

$$\frac{7}{8}k = E[Z] = \sum_{j<7k/8} j \cdot p_j + \sum_{j \geq 7k/8} j \cdot p_j$$

$$\leq \frac{7k-1}{8} \sum_{j<7k/8} p_j + k \sum_{j \geq 7k/8} p_j = \frac{7k-1}{8}(1-p) + kp \leq \frac{7k-1}{8} + kp$$

$$kp \geq \frac{7}{8}k - \frac{7k-1}{8} = \frac{1}{8} \implies p \geq \frac{1}{8k}$$

The algorithm used to solve MAX-3SAT with this probability is called **Johnson's algorithm**.

the algorithm: repeatedly generate random assignments until one satisfies $\geq \frac{7}{8}k$ clauses.

- it's a 7/8-approximation algorithm for MAX-3SAT. (N.B.: Håstad proved that no α-approximation exists for $\alpha > 7/8$ unless P=NP).
- as seen above, each iteration succeeds with probability $\geq \frac{1}{8k}$; using the waiting-time bound for

a first success, the expected number of trials is $\leq 8k$.

- the total expected runtime is $E[T] = O(8k) \cdot O(n+k) = O(k^2 + kn)$

> This process turns a Monte Carlo algorithm into a Las Vegas algorithm (it always produces a correct $7/8$ approximation, but runtime is an expectation).

4.5 | Tail Inequalities

Expected runtimes describe average behavior, but tail inequalities provide bounds on the probability that an algorithm's performance strays far from its expectation.

Markov's Inequality

Let X be a non-negative random variable ($X \geq 0$) with expectation $E[X]$.

For any $t > 0$:

$$Pr[X \geq t] \leq \frac{E[X]}{t} \quad \text{or with } t = k \cdot E[X] : \quad Pr[X \geq k \cdot E[X]] \leq \frac{1}{k}$$

- requires the least information (only $E[X]$ and non-negativity), making it universally applicable but providing the weakest (slowest decay) bound

Chebyshev's Inequality

Let X be a random variable with expectation $E[X]$ and standard deviation $\sigma[X] = \sqrt{E[(X - E[X])^2]}$. For any $t > 0$:

$$Pr[|X - E[X]| \geq t \cdot \sigma[X]] \leq \frac{1}{t^2}$$

- incorporates variance to measure spread around the mean; the deviation probability drops quadratically ($1/t^2$)

Chernoff Bounds

Let $X = \sum_{i=1}^{n} X_i$ be the sum of *independent 0–1 random variables*.

For any $\mu \geq E[X]$:

above the mean (for any $\delta > 0$):

$$Pr[X > (1+\delta)\mu] < \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}} \right)^{\mu}$$

below the mean (for $0 < \delta < 1$):

$$Pr[X < (1-\delta)\mu] < e^{-\delta^2 \mu / 2}$$

- the sum of independent 0-1 random variables is exponentially tightly concentrated around its mean

Example: Chernoff bounds on coin flips
Consider n flips of a fair coin, where X is the total number of heads.

Here, $\mu = E[X] = \frac{n}{2}$.

We want the probability of getting more than $\frac{3}{4}n$ heads (i.e., more than 75% heads).

We set the Chernoff threshold $(1+\delta)\mu$ equal to $\frac{3}{4}n$:

$$(1+\delta)\cdot E[X] = \frac{3}{4}n \iff (1+\delta)\left(\frac{n}{2}\right) = \frac{3}{4}n$$

$$1+\delta = \frac{3}{2} = 1.5 \implies \delta = 0.5$$

So, we have that

$$Pr\left[X > (1+0.5)E[X]\right] = Pr\left[X > \frac{3}{4}n\right] < \left(\frac{\sqrt{e}}{1.5^{1.5}}\right)^{n/2} \approx 0.9^{n/2}$$

If we compare the results with the actual probabilities:

- $n = 2$: exact $Pr\left[X > \frac{3}{4}n\right] = 0.25 < 0.9$
- $n = 3$: exact $Pr\left[X > \frac{3}{4}n\right] = 0.125 < 0.86$
- $n = 4$: exact $Pr\left[X > \frac{3}{4}n\right] = 0.0625 < 0.81$
- $n = 10$: exact $Pr\left[X > \frac{3}{4}n\right] = 0.0546875 < 0.59$
- $n = 100$: exact $Pr\left[X > \frac{3}{4}n\right] \approx 2.82 \cdot 10^{-7} < 0.0052$

Chernoff bounds overestimate the true error probability significantly, but they are widely used because they are simple to compute and guarantee exponential decay wrt n.

4.6 | Key Takeaways

The Probabilistic Method

Non-constructive proof technique used to prove the **existence of a specific solution or structure**. It can be done:

- **via expectation:** if you can show that the expected value of a random variable is $E[X] = \mu$, then there mathematically must exist at least one specific outcome in the sample space that achieves $\geq \mu$ (and analogously, one that achieves $\leq \mu$)
- **via probability:** if you can prove that the probability of randomly generating a valid configuration is greater than zero ($Pr[\mathcal{E}] > 0$), then that configuration definitively exists

Converting Monte Carlo to Las Vegas

A Monte Carlo algorithm (fast, but potentially returns an incorrect output) can be upgraded into a Las Vegas algorithm (always correct, but with an expected, non-guaranteed runtime) if and only if you can efficiently verify the correctness of the output:

- wrap the Monte Carlo algorithm in a loop that repeats until the verification step succeeds

Analyzing Algorithm Probability

To analyze the expected runtime or an expected score of a randomized algorithm:

1 | **define indicator variables:** break the total cost down into binary events.

define an indicator variable $X_i \in \{0, 1\}$ (e.g., $X_{i,j} = 1$ if a comparison happens between i and j, or $Z_j = 1$ if clause j is satisfied).

2 | **calculate individual probability:** compute the probability that a single indicator evaluates to 1 ($Pr[X_i = 1]$)

because it is a binary 0-1 variable, its expected value is exactly its probability: $E[X_i] = Pr[X_i = 1]$

3 | **apply linearity of expectation:** use $E[\sum X_i] = \sum E[X_i]$ to add up all the individual probabilities, getting the overall expected total cost

Choosing a Tail Inequality

Tail inequalities quantify concentration by bounding the probability that a random variable deviates far from its mean. We can choose the bound based on what information is available:

- **markov's inequality:** use when $X \geq 0$ and you *only* know $E[X]$.

  weakest bound; probability drops off polynomially ($O(1/k)$).

- **chebyshev's inequality:** use when you know both $E[X]$ and the variance $\sigma^2[X]$ (or standard deviation σ).

  wtronger bound; probability drops off quadratically ($O(1/t^2)$).

- **chernoff bounds:** use when $X = \sum X_i$ is a sum of *independent 0–1* (Bernoulli) random variables.

  wtrongest bound; probability drops off exponentially.

Summary of Tail Inequalities

1 | **Markov:** $Pr[X \geq k \cdot E[X]] \leq \frac{1}{k}$

assumptions: $X \geq 0$, $k > 0$.

2 | **Chebyshev:** $Pr[|X - E[X]| \geq t \cdot \sigma[X]] \leq \frac{1}{t^2}$

assumptions: Known variance $\sigma^2[X]$.

3 | **Chernoff (Above Mean):** $Pr[X > (1+\delta)\mu] < \left(\frac{e^\delta}{(1+\delta)^{1+\delta}}\right)^\mu$

assumptions: $X = \sum_{i=1}^{n} X_i$ with independent $X_i \in \{0, 1\}$ and $\mu \geq E[X]$.

4 | **Chernoff (Below Mean):** $Pr[X < (1-\delta)\mu] < e^{-\delta^2\mu/2}$

Assumptions: Same as above, with $0 < \delta < 1$.

CHAPTER 5

MILP

5.1 | Linear Programming

The objective of linear programming is to maximise (/minimise) a linear function in presence of linear (in)equality constraints.

> Example
>
> We could, for example, want to maximise a function like
>
> $$3x_1 - x_2 - 2x_3$$
>
> subjects to constraints like:
>
> - $4x_2 - 0.5x_3 \leq 10$
> - $x_1 \geq 1$
> - $x_2 + 2x_3 \leq -2$
> - $x_1 + 2x_2 + 7.7x_3 = -9$

(euclidean norm - modulo di un vettore, square root of the sum of the squared components)

> Def. 20 **Linear Program**
>
> A linear program is defined by
>
> - $A = (a_{ij})_{ij} \in \mathbb{R}^{m \times n}$ (constraint matrix)
> - $c \in \mathbb{R}^n$ (objective function coefficients)
> - $b \in \mathbb{R}^m$ (bounding limits)
> - $x \in \mathbb{R}^n$ (decision variables)
>
> We want to maximize the **objective function**
>
> $$c^T x = c_1 x_1 + \cdots + c_n x_n$$
>
> subject to
>
> $$\begin{aligned} a_{i,1}x_1 + \cdots + a_{i,n}x_n &\leq b_i \quad \text{for } i \in I_- \\ a_{i,1}x_1 + \cdots + a_{i,n}x_n &\geq b_i \quad \text{for } i \in I_+ \\ a_{i,1}x_1 + \cdots + a_{i,n}x_n &= b_i \quad \text{for } i \in I_0 \end{aligned}$$

The system of constraints can be visualized as a matrix-vector multiplication $Ax \leq b$:

$$\underbrace{\begin{pmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \dots & a_{m,n} \end{pmatrix}}_{\text{matrix } A \text{ (coefficients)}} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\text{vector } x \text{ (variables)}} \begin{matrix} \leq \\ \leq \\ \vdots \\ \leq \end{matrix} \underbrace{\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}}_{\text{vector } b \text{ (bounds)}}$$

The objective function uses a separate vector c (the coefficients/weights) multiplied by our variable vector x. We want to find the specific x that maximizes this result:

$$\text{Maximize} \quad \underbrace{\begin{pmatrix} c_1 & c_2 & \dots & c_n \end{pmatrix}}_{\text{row vector } c^T \text{ (weights)}} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\text{column vector } x \text{ (variables)}} = \text{total value}$$

> (so, the matrix multiplication dictates which values are “legal” for x - the so-called **feasible region** -, and, for any legal value, the vector operation can assign it its “value”)
>
> (therefore, the point is to search through the valid x factors and find which one maximises/minimises the vector product with c)

So, we have that:

- $x \in \mathbb{R}^n$ is a *feasible solution* if the constraints apply
- the **feasible region** is the set of all feasible solutions
- a maximiser is an (optimal) feasible solution, and its maximum value is called the **optimum**

We work, WLOG, under the assumption that all our inequalities are in the $\leq$ form (i.e. $I_+ = I_0 = \varnothing$).

Therefore, our problem becomes to maximise $c^T x$ subject to $(a_{i,1}, a_{i,2}, \dots, a_{i,n})x \leq b_i$, with $x \geq 0$.

(which can also be expressed, in a shorter form, as

$$\max(c^T x : x \in \mathbb{R}^n,\ Ax \leq b,\ x >= 0)$$

)

Standard Form

The standard form (used for simplex algorothms) is:

- maximising $c^T x$
- subject to $Ax = b$
- with $x \geq 0$

> I
>
> f we have equations of a different form, we can convert them into standard form as follows:

$$a_{i,1}x_1 + \cdots + a_{i,n}x_n \leq b \implies a_{i,1}x_1 + \cdots + a_{i,n}x_n + s_i = b$$

where $s_i \geq 0$ is called *slack*

$$a_{i,1}x_1 + \cdots + a_{i,n}x_n \geq b \implies a_{i,1}x_1 + \cdots + a_{i,n}x_n - s_i = b$$

where $s_i \geq 0$ is called *surplus*

Example

naïve form

$$\begin{aligned} \max \quad & 3x_1 - x_2 - 2x_3 \\ \text{subject to} & \\ 4x_2 - 0.5x_3 &\leq 10 \\ x_1 &\geq 1 \\ x_2 + 2x_3 &\leq -2 \\ x_1 + 2x_2 + 7.7x_3 &= -9 \\ x_1, x_2, x_3 &\in \mathbb{R} \end{aligned}$$

standard form

$$\begin{aligned} \max \quad & 3x_1 - (x_2^+ - x_2^-) - 2(x_3^+ - x_3^-) \\ \text{subject to} & \\ 4(x_2^+ - x_2^-) - 0.5(x_3^+ - x_3^-) + s_1 &= 10 \\ x_1 - s_2 &= 1 \\ (x_2^+ - x_2^-) + 2(x_3^+ - x_3^-) + s_3 &= -2 \\ x_1 + 2(x_2^+ - x_2^-) + 7.7(x_3^+ - x_3^-) &= -9 \\ x_1, x_2^+, x_2^-, x_3^+, x_3^-, s_1, s_2, s_3 &\geq 0 \end{aligned}$$

5.1.1 | Polyhedral geometry

Recalls

Let us recall some algebraic concepts.

Firstly, the **inner product** (or dot product if in Euclidean space) is an operation that combines two vectors to produce a scalar, measuring their similarity.

It can be seen in two different ways:

- algebraically

$$\sum_{i=1}^{n} v_i w_i$$

  where we take the n-th coordinate of v and multiply it by the n-th coordinate of w, and add them all up

- geometrically

$$\cos \sphericalangle(v, w)|v||w|$$

  $||v||$ and $||w||$ are the magnitudes/lengths of the vectors, and $\sphericalangle$ is the angle between v and w (of which we take the cosine)

  Furthermore, we have that for a vector $a \in \mathbb{R}^n \setminus \{0\}$ and $b \in \mathbb{R}$:

  - a set $\{x \in \mathbb{R}^n \mid \sum_{i=1}^{n} a_i x_i = b\}$ is a **hyperplane** (a single row from a $Ax = b$ matrix)
    a hyperplane is meant to represent a flat, uncurved surface; in 2D, a it's a straight line, in 3D it's a flat plane, in 4D and on it's represented by the set above

- changing the equal sign into an inequality ($\{x \in \mathbb{R}^n \mid \sum_{i=1}^{n} a_i x_i \leq b\}$) yields a **half-space**: the hyperplane slices the space in two, and the halfspace is everything on one side of tue cut (keeping the boundary line)

(the a above is a **normal vector**, which is essentially a vector of coefficients for the xs; in general, a normal vector is one that is perpendicular)

We have that a normal vector a points to the "upper half space" H_+ of a hyperplane H (since it's perpendicularm it acts as an arrow pointing away from the boundary, indicating which half of the space is "positive").

This way, the **feasible region** P of an LP ($\max\{c^T x : Ax \leq b,\ x \geq 0\}$) can be seen as an *intersection of $m + n$ halfspaces*:

- the matrix has m rows, so m constraint inequalities forming m halfspaces
- the standard form also includes the positive orthant rule $x \geq 0$, which generates n boundary inequalities ($x_1 \geq 0$, $x_2 \geq 0$ etc:w)

Def. 21 **Polyhedron, Polytope**

A **polyhedron** is a set $M \subseteq \mathbb{R}^n$ that is the intersection of finitely many halfspaces of $\mathbb{R}^n$.

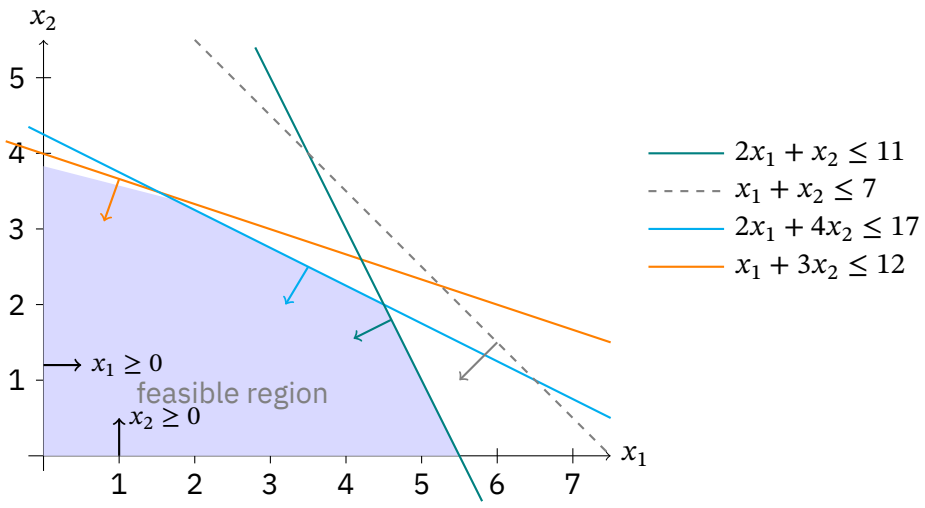
- it is built from straight cuts, so it has flat faces and sharp corners; it also can be completely closed off, or it can be open and stretch out into infinity

A **polytope** M is a *bounded polyhedron* (it does not stretch to infinity): it is contained in some fixed-size n-dimensional ball centered at $0 \in \mathbb{R}^n$

I.e., $\sup_{x \in M} |x| < \infty$:

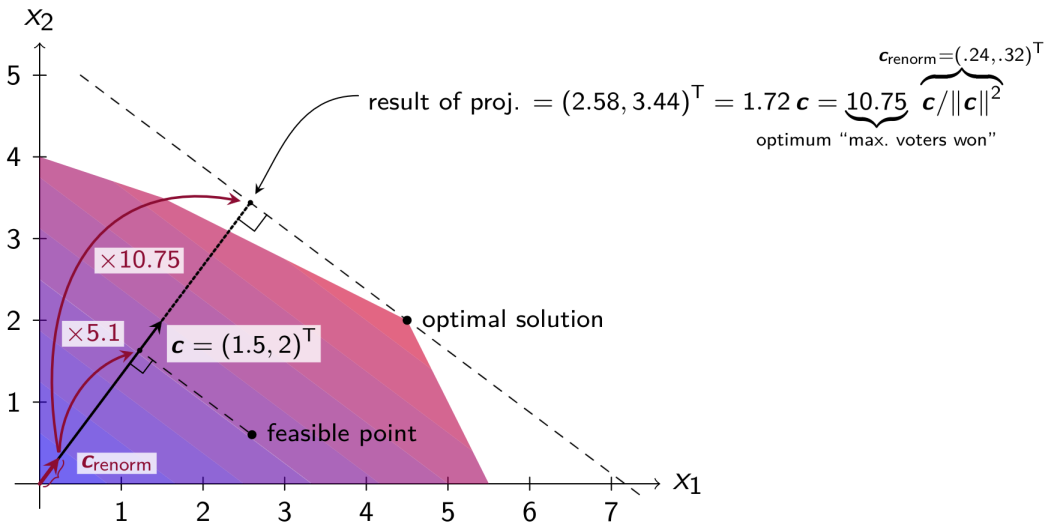
- this means that the maximum distance from the origin is a finite number (the length of x, $|x|$ (its distance from the origin 0)

We can visualise the example we mentioned before on a 2D plane:



- note that the constraint $x_1 + x_2 \leq 7$ is "redundant" (it is "dominated"(? shaded?) by the other stronger constraints)

If we consider this geometrically:



We can see that:

- note: a **linear hull** (or span) of a vector c, $[c]$, is the infinite line drawn through the origin that follows c's direction forever
- the vector $c = (1.5, 2)^T$ is our objective function, and our goal (in a maximization problem) isto find the point in the feasible (purple) region that extends the furthest in the direction of c
- for every feasible point, we can consider how "good" it is, by drawing a 90-deg line from the dot onto c - the place where it lands is its "score"

So, we want to see how far a feasible point v extends in the direction of our objective vector c.

Algebraically, the point's "score" is the dot product $c^T v$. Geometrically, this score is found by orthogonally projecting v onto the line of c. However, the standard vector projection formula (onto the linear hull $[c]$) is a bit complicated:

$$\text{proj}_{[c]} v = (c^T v) \cdot \left(\frac{c}{\|c\|^2} \right)$$

But we can introduce a custom vector:

$$c_{\text{renorm}} = \frac{c}{\|c\|^2}$$

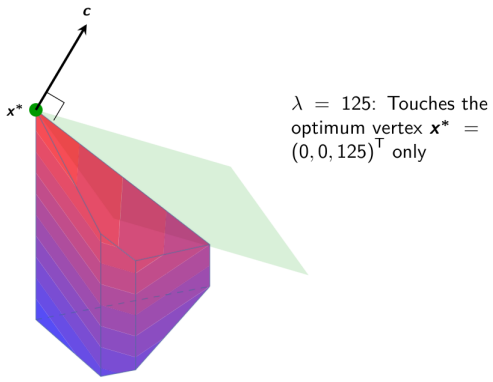
(c_{renorm} is like a custom "measuring stick" for which one physical step along the line equals exactly one unit of profit in the score).

Our projection formula is now:

$$\text{proj}_{[c]} v = \lambda \cdot c_{\text{renorm}} \quad \text{where } \lambda = c^T v$$

(if a point v has a profit score of 5.1, its geometric projection drops onto the objective line exactly 5.1 "measuring sticks" away from the origin;

This way, the optimal solution is the point that allows us to maximise the number of c_{renorm} without slipping outside the feasible region.



If we move from 2D to 3D, our feasible region is a solid block: to evaluate it, we don't use a line, but rather a 2D sheet (green hyperplane in the figure).

- $\{x : c^T x = \lambda\}$ represents the huperplane, and it's a "level set": a sheet where every single point produces the same profit score λ (it is perpendicular to c)
- to maximise profit, we want λ to be as large as possible, which means sliding the hyperplane up along the c arrow: the last point of contact between the green hyperplane and the purple shape is our maximum

5.1.2 | Convexity

Def. 22 Convexity

A shape M is convex if you can pick any two points x and y inside it and draw a line between them that stays completely inside the shape.

This is expressed by

$$(1-\lambda)x + \lambda y$$

λ ranges from 0 to 1, where $\lambda = 0$ gives x, $\lambda = 1$ gives y, and the other values are the points in-between (e.g. $\lambda = 0.5$ is the midpoint).

A function f is convex if, when you pick any two points on its curve and draw a straight line between them, the curve will always sag below or equal to that straight line.

The line is expressed by:

$$f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y)$$

where

- $f((1-\lambda)x + \lambda y)$ is the height of the function curve betweeen x and z
- $(1-\lambda)f(x) + \lambda f(y)$ is the height of the straight line connecting the two points
- (so, by $\leq$, the curve needs to sit below the line)

We have that convex sets are stable (/closed) under intersection. This tells us that, since a halfspace is a convex set, a polyhedron, which is built by overlapping halfspaces, will also be convex.

We also have that convex functions don't have *local minima* (any local minimum is a global minimum).

Any affine function (function consisting of a linear transformation followed by a translation/offset (e.g. $y = mx + b$)) is both convex and concave (because it has no curvature).

Thm. 7 **[Maximizing Convex Functions]**

A non-constant continuous convex function $f : M \to \mathbb{R}$ defined on a convex closed set $\varnothing \neq M \subseteq \mathbb{R}^n$ is either unbound from above ($\sup_M f = +\infty$) or, provided it has a maximum (it is a polytope), *every maximiser* is at the boundary of M.

(this means optimal solutions are always found on the edges/vertices, never inside the feasible region)

Geometrically, a linear objective function behaves like a rigid uniformly tilted plane, while the convex feasible region M acts as a "fenced-in" spatial domain. To maximize the objective function, we need to go up the upward slope of the tilted plane. Because linear and affine functions have no curvature, the ascent is uninterrupted, so we keep going up indefinitely until we are obstructed by the outer "fence" of the feasible domain. Consequently, the point of highest elevation (the optimal solution) cannot exist in the interior; the lack of curvature forces the maximizer to be on the boundary of M.

Some special cases are:

- if $M = \varnothing$ (no valid region), convention dictates $\max f = -\infty$
- if the objective function aligns perfectly parallel to a boundary face, there are infinitely many maximizers
- if the objective function pushes into a single extreme point (vertex), the maximizer is unique
- even if M stretches to infinity in one direction, a finite maximum can still exist if the objective function points away from the infinite direction

N.B.: the **maximum** is the maximum value, and it is unique, while the **maximiser** is the optimal solution that generates the maximum (the xs), and there can be more than one

Def. 23 **Convex Hull**

Given $M \subseteq \mathbb{R}^n$ and a set of points $(v^i)_{i \leq n}$, the **convex hull** is the smallest possible convex shape that completely wraps up all of the points.

(algebraically, it's the set of all points x that can be formed by a *convex combination* of a finite number of points from M.)

Formally, it is the set of all x for which there are

- $N \geq 1$ points $v^1, \dots, v^n \in M$
- scalars $\lambda_1, \dots, \lambda_n \in [0, 1]$ s.t. $\sum_{i=1}^{N} \lambda_i = 1$ (called **weights**)

such that

$$x = \sum_{i=1}^{N} \lambda_i v^i$$

(as mentioned above, this is referred to as the **convex combination**).

(the sum is a weighted average, and, since every weight is positive and they all add up to 1, we know that the new point x will land inside the boundary of the v^i)

Thm. 8 **[Finite basis theorem for polytopes]**

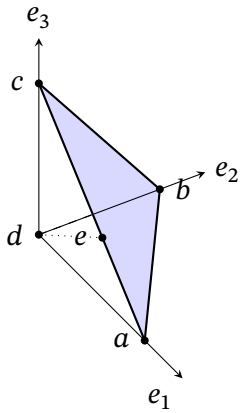
A set of points is a polytope (a bounded polyhedron) if and only if it is the convex hull of a finite number of points (its vertices).

> (before, we had defined a polytope "from outside in", as a bounded shape generated by intersecting flat halfspace constraints
>
> now, we see that we can also define it "from the inside out": instead of listing the boundary lines, we list its corners and take their convex hull)

Def. 24 **Extreme points**

For a set $M \subseteq \mathbb{R}^n$, a point $x \in M$ is called an **extreme point** if it cannot be expressed as a convex combination of $y, z \in M \setminus \{x\}$

(it forms a convex corner)



$$a = \begin{pmatrix}1\\0\\0\end{pmatrix}, \quad b = \begin{pmatrix}0\\1\\0\end{pmatrix}, \quad c = \begin{pmatrix}0\\0\\1\end{pmatrix},$$

$$d = \begin{pmatrix}0\\0\\0\end{pmatrix}, \quad e = \begin{pmatrix}0.6\\0\\0.4\end{pmatrix}$$

Figure 5.1: e is not extreme, but a, b, c are; the surface equals $\text{conv}\{a, b, c\}$.

Thm. 9 **[(Extreme points)]**

For a polyhedron $\varnothing \neq P = \{x | Ax \leq b\}$, these assertions are equivalent:

- P has at least one extreme point
- P does not contain a line (an infinite line that extends infinitely in both directions)

  i.e. $\forall y \in \mathbb{R}^n, v \in \mathbb{R}^n \setminus \{0\}, \ \{y + \lambda v | \lambda \in \mathbb{R}\} \nsubseteq P$

  (you start at a point y inside the shape, pick a direction v and use the multiplier λ (any number in $[-\infty, +\infty]$) to extend it forever in both directions; if this can be done without exiting the shape, it "contains" the line)
- the set of rows $A = \{a_{i,1} \dots, a_{i,n}\}$ contains n linearly independent $1 \times n$ row vectors

  linearly dependent rows are parallel lines (derivable from one another with linear transformations), while linearly independent rows will have different angles and intersect, creating an angle; here, we're in an n-dimensional space, so, to obtain a corner, we need n different rows intersecting (and since the rows intersect from every dimension, there is no "free" dimension for an infinite line to run tthrough)

We said (in thm 5.1.2) that the solutions must be on the edges of our feasible region.

We want to be more specific, and to prove that they are on *extreme points*.

> Thm. 10 **[Bauer's Maximum Principle]**
>
> When maximising c^T over $P = \{x : Ax \leq b\}$, suppose there exists at least one extreme point. Then, either $x \mapsto c^T x$ is unbounded on P (max $= \infty$), or there exists an optimal solution x_* among the extreme points of P.

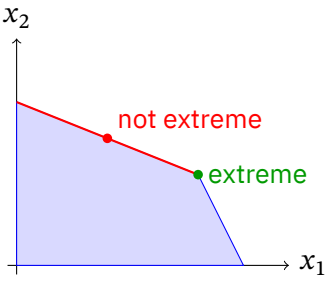
- since our formulation $\max\{c^T x | Ax \leq b,\ x \geq 0\}$ has a feasible region in $x \geq 0$, unless the polyhedron is empty, extreme points exist!

> Thm. 11 **[Finiteness of extreme points]**
>
> In a polyhedron, there can only be finitely many extreme points.

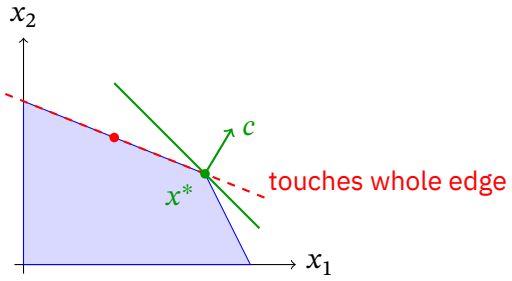
- but they can still be a lot!

There are actually three ways to define extreme points!

Extreme point (inner view): $x \in P$ is no convex combination of two points $y, z \in P \setminus \{x\}$.



Vertex (outer view): there is c such that x is the *unique* maximizer of $c^\mathsf{T} x$ over P



(a point is a vertex if you can tilt a board (the green line) so that it touches that one specific point and nothing else)

And! both notions also have an *algebraic* characterization:

> Def. 25 **Basic feasible solution**
>
> $x \in P$ is a **basic feasible solution** if, among the constraints $\sum_{j=1}^{n} a_{i,j} x_j \leq b_i$ that are *active* at x (i.e., hold with equality (so, the points on the boundary)), there are n with linearly independent rows $(a_{i,1}, \ldots, a_{i,n})$

(we need to be touching 'walls' for all dimensions) (the only way to stand against n different, non-parallel walls at the same itme is to be on the corner where they intersect)

> Thm. 12 **[Bertismas, Tsitsklis, 1997]**
>
> (for a polyhedron P) x is an extreme point of $P \iff x$ is a vertex of P

5.2 | Duality

Thm. 13 **[Farkas' Lemma, 1894]**

These two statements are equivalent:

- there exists $x \geq 0 \in \mathbb{R}^n$, with $Ax \leq b$
  (there is a valid solution)
- if, for each vector $y \in \mathbb{R}^m$, $y \geq 0$, we have $A^T y \geq 0$, then we must also have $y^T b \geq 0$
  - we introduce a new vector y in the m-dimensional space of the rows (constraints), which acts as a set of *weights* for the constraints
  - we check for contradiction: we take all the constraint equations and multiply each one by one by a positive weight and add them up
    - $A^T y \geq 0$: the weights are s.t. the left side is positive
    - $y^T b \geq 0$: the right side (the bounding) is also positive
    - (if we made the left side positive, but the right side somehow negative, the system would be claiming that a positive value is $\leq$ a negative value)
      so this basically says: if the left side is positive, the right side must also be positive (and there are no contradictions)

With this, we see that solution existence in Euclidean n-dimensional space can be translatedd in a universally quantified statement in an m-dimensional space.

> Every Linear Program (the **primal**) has a **dual**. The Primal seeks to maximize profit given resource limits, while the Dual seeks to minimize the cost (or weights) of those resources given minimum requirements.

We have two forms of duality.

5.2.1 | Weak Duality

This proves that the Dual problem acts as an upper bound for the Primal problem:

primal problem (here $x \in \mathbb{R}^n$)

$$\max\{c^T x : Ax \leq b, x \geq 0\}$$

(maximize $c^T x$ subject to resource limits $Ax \leq b$)

dual problem (here $y \in \mathbb{R}^m$)

$$\min\{b^T y : A^T y \geq c, y \geq 0\}$$

(minimize $b^T y$ subject to minimum requirements $A^T y \geq c$)

We can lik any feasible solution x of the primal and y of the dual through a chain of inequalities:

$$c^T x = x^T c \leq x^T A^T y = y^T Ax = (Ax)^T y \leq b^T y$$

And looking at the start and end of the chain:

$$\max_{Ax \leq b, x \geq 0} c^T x \leq \min_{A^T y \geq c, y \geq 0} b^T y$$

This essentially tells us that the highest profit the primal can ever achieve is always trapped beneath or equal to the lowest cost the dual can find.

(therefore, if a primal is unbounded to $+\infty$, the dual's feasible region must be empty, i.e., $+\infty$).

5.2.2 | Strong Duality

Strong duality is the guarantee that if both the primal and the dual have a finite optimum (neither is unbounded or empty), their optima coincide.

The highest possible profit exactly equals the lowest possible cost:

$$\max_{Ax \leq b, x \geq 0} c^T x = \min_{A^T y \geq c, y \geq 0} b^T y$$

The formulations above is also called **symmetric** duality, where both problems have non-negativity constraints ($x \geq 0$ and $y \geq 0$).

If the primal drops its $x \geq 0$ rule (i.e. variables can be negative), the dual can compensate by tightening its constraints from an inequality to a strict equality.

This is **asymmetric** duality, and strong duality still holds:

$$\max \{c^T x : Ax \leq b\} = \min \{b^T y : A^T y = c, y \geq 0\}$$

LP Duality Conversion

When converting a general Linear Program with mixed constraints and variable bounds to its dual, we use the rules below:

primal problem (maximize)	**dual problem (minimize)**
$\max c^T x$	$\min b^T y$
s.t. $(a_{i,1}, \dots, a_{i,n})x \leq b_i, \quad i \in I_-$	s.t. $y_i \geq 0, \quad i \in I_-$
$(a_{i,1}, \dots, a_{i,n})x \geq b_i, \quad i \in I_+$	$y_i \leq 0, \quad i \in I_+$
$(a_{i,1}, \dots, a_{i,n})x = b_i, \quad i \in I_0$	y_i unrestricted, $\quad i \in I_0$
$x_j \leq 0, \quad j \in J_-$	$(a_{1,j}, \dots, a_{m,j})^T y \leq c_j, \quad j \in J_-$
$x_j \geq 0, \quad j \in J_+$	$(a_{1,j}, \dots, a_{m,j})^T y \geq c_j, \quad j \in J_+$
x_j unrestricted, $\quad j \in J_*$	$(a_{1,j}, \dots, a_{m,j})^T y = c_j, \quad j \in J_*$

- a tight constraint (=) in one domain creates an unrestricted variable in the other
- a standard inequality ($\leq$ in max / $\geq$ in min with ≥ 0 variables) maps to a non-negative dual counter-part

Often, we encounter minimization problems instead of maximization ones. The principle of duality also applies to these.

In general, we have that:

- the primal bound tells us what solution quality can guaranteedly be achieved (certificate)

- the dual bound gives us guaranteed limitations for optimality (feasibility will not be possible after violating this limit)

To calculate whether a pair of primal and dual solutions is optimal, we can use *complementary slackness*.

slack

In any inequality constraint like $a_i x \leq b_i$, *slack* is the leftover amount of a resource: $\text{slack}_i = b_i - a_i x$.

- if $\text{slack}_i = 0$, the constraint holds with exact equality ($a_i x = b_i$) - the constraint is active or binding (100% of the resource was used)
- if $\text{slack}_i > 0$, we have leftover capacity (the constraint is inactive)

Thm. 14 **[Complementary Slackness]**

Let $x^* \in \mathbb{R}^n$ and $y^* \in \mathbb{R}^m$ be feasible solutions for the primal LP and dual LP respectively. The following three statements are equivalent:

1 | x^* and y^* are optimal solutions

2 | $c^T x^* = b^T y^*$ (strong duality holds)

3 | complementary slackness holds for all $i = 1, \dots, m$ and $j = 1, \dots, n$:

$$y_i^* \cdot \left(\sum_{j=1}^{n} a_{i,j} x_j^* - b_i \right) = 0 \quad \forall i = 1, \dots, m$$

$$x_j^* \cdot \left(\sum_{i=1}^{m} a_{i,j} y_i^* - c_j \right) = 0 \quad \forall j = 1, \dots, n$$

Essentially, for any pair of primal/dual variables and constraints, at least one side of the pair must have zero slack:

- if a primal constraint has leftover capacity ($\text{slack}_i > 0$), its dual multiplier y_i^* must be 0.
- if a dual variable is positive ($y_i^* > 0$), the corresponding primal constraint must be active ($\text{slack}_i = 0$).
- if a decision variable is produced ($x_j^* > 0$), its corresponding dual constraint must hold with exact equality

5.2.2.1 Modeling the Maximum Flow Problem

Flow

Given a directed graph $G = (V, E)$, we designate two nodes:

- a *source* s (where flow originates), with no incoming edges: $N^-(s) = \varnothing$
- a *sink* t (where flow is consumed), with no outgoing edges: $N^+(t) = \varnothing$

Each edge $e \in E$ has a non-negative *capacity* $c_e \geq 0$, representing the maximum amount of flow it can handle.

A **flow** is an assignment of values f_e to each edge $e \in E$ such that two laws hold:

1. *capacity constraint*: the flow on an edge cannot exceed its capacity
2. *flow conservation*: for every node except the source and sink, the total flow entering the node must exactly equal the total flow leaving it (what goes in must come out)

The **Maximum Flow Problem** asks: what is the maximum amount of total flow we can push from the source s to the sink t, while respecting all capacities and conservation rules?

Problem requirement

1 | digraph $G = (V, E)$, source s, i.e., $N^-(s) = \emptyset$, sink t, i.e., $N^+(t) = \emptyset$

2 | capacity $E \ni e \mapsto c_e \in \mathbb{R}_{\geq 0}$

3 | flow on each edge

$$E \ni e \mapsto f_e \in \mathbb{R}_{\geq 0}$$

4 | flow conservation, except for s and t
(what enters an intermediate node must exactly equal what exits it)

5 | find flow f maximizing the cumulative flow leaving the source

LP translation

1 | **(problem-specific constants)**
the vertex set V and edge set E determine the dimensions of our LP matrices; the source has no incoming edges ($N^-(s) = \emptyset$) and the sink has no outgoing edges

2 | **(problem-specific constants)**
the capacities c_e form the right-hand bound vector b_{cap} for our inequality constraints

3 | **LP decision vars:**

$$f_e \in \mathbb{R}, \quad 0 \leq f_e \leq c_e \; \forall e \in E$$

(we map the flow on each edge to a column vector $x = [f_1, \dots, f_m]^T$; the inequality $Ax \leq b$ is formed here as $Ix \leq b_{cap}$, where the identity matrix I isolates and limits each edge individually)

4 |

$$\sum_{v \in N^-(w)} f_{vw} = \sum_{v \in N^+(w)} f_{wv} \quad \forall w \in V \setminus \{s, t\}$$

- the left sum $\sum_{v \in N^-(w)}$ calculates all flow coming **into** node w from its in-neighbors
- the right sum $\sum_{v \in N^+(w)}$ calculates all flow going **out** of node w to its out-neighbors
- this forms the equality block $A_{cons}x = 0$. A_{cons} is the **node-edge incidence matrix**: one row per intermediate node, one column per edge; edges entering the node get $+1$, exiting get -1, and unattached get 0; the bound vector is 0 because no flow can accumulate

5 | **maximize** $\sum_{v \in N^+(s)} f_{sv}$

- to map this to the standard max $c^T x$, the cost vector c acts as a (binary) filter
- for every edge leaving the source s, its entry in c is 1, for all other edges, it is 0
- the dot product $c^T x$ therefore isolates and sums only the variables f_{sv}, ignoring the rest of the network entirely

5.3 | Algorithms and Complexity

The Simplex Algorithm

The Simplex algorithm executes a gradient ascent by navigating along the edges (the 1-skeleton) of a feasible polyhedron, moving from one vertex to an adjacent vertex to increase the objective score.

It is highly effective in practice, but it can be disrupted by geometrical problems and matrix degenerate states.

- an n-dimensional vertex is geometrically defined by the intersection of n independent boundary hyperplanes; a vertex is **degenerate** when more than n constraints cross at the exact same coordinate point (an over-determined corner)
- because simplex tracks vertices using algebraic combinations of n active constraints (a basis), a degenerate corner possesses multiple valid bases describing the same position; when Simplex pivots at a degenerate vertex, it swaps constraints in its matrix without moving physically in space (this results in a physical step size of $\Delta x = 0$ and zero gain)
- because a zero-step pivot fails to increase the objective score, the algorithm loses its progress mechanism; without a strict tie-breaking rule, Simplex can endlessly cycle through the equivalent algebraic bases of that single degenerate point in an infinite loop

To prevent cycling, **Bland's Rule** enforces a deterministic hierarchy for selecting pivot variables:

- if multiple variables qualify to enter or leave the basis, always choose x_k with the minimum index k

by prioritizing smaller variable indices, Bland's Rule breaks symmetry and guarantees that Simplex cannot re-enter a previously visited basis, forcing the algorithm to break out of degenerate vertices in a finite number of steps

- Simplex performs extremely well in practice, typically taking a polynomial (expected) number of iterations, with worst-case $\mathcal{O}(2^n)$

Alternative LP Solvers

Ellipsoid Method

- proved LP $\in$ P (polynomial-time solvable)
- shrinks bounding ellipsoids around the feasible region
- solves LPs with exponentially many constraints if an efficient oracle identifies violated cuts
- impractical due to high numerical instability and slow convergence

Interior Point Method

- both polynomial-time theoretically and fast in practice
- travels through the interior of the feasible region along the "Central Path"
- extends to non-linear/non-convex optimization, semi-definite programming (SDP), and MILP branch-and-bound solvers